

PCS 224

Semiconductor Physics

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Mobina

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Contents

1	Formula Sheet	2
2	Tutorials	4
2.1	Tutorial 1 - PART 1	4
2.2	Tutorial 1 Solutions - PART 1	5
2.3	Tutorial 2 - PART 1	6
2.4	Tutorial 2 Solutions - PART 1	7
2.5	Tutorial 3 - PART 1	8
2.6	Tutorial 3 Solutions - PART 1	8
2.7	Tutorial 3 - PART 2	11
2.8	Tutorial 3 Solutions - PART 2	13
2.9	Tutorial 3 - PART 3	17
2.10	Tutorial 3 Solutions - PART 3	18
3	Practice Problems	20
3.1	Set 6	20

Formula Sheet

Photon Energy and Waves

$$f\lambda = c$$

$$E = hf = \frac{hc}{\lambda}$$

$$hf = K_{\max} + \phi$$

$$K_{\max} = eV_{\text{stop}}$$

$$E_n = -\frac{13.6}{n^2} \text{ eV}$$

Current Density and Conductivity

$$I = \frac{\Delta Q}{\Delta t}$$

$$I = nqAv_d$$

$$J = \frac{I}{A}$$

$$\vec{J} = \sigma \vec{E}$$

$$\sigma = \frac{1}{\rho} = e(n\mu_n + p\mu_p)$$

Carrier Concentrations

$$n_0 = N_C e^{-\frac{E_C - E_F}{k_B T}}$$

$$p_0 = N_V e^{-\frac{E_F - E_V}{k_B T}}$$

$$n_0 p_0 = n_i^2$$

$$n_i^2 = N_C N_V e^{-\frac{E_g}{k_B T}}$$

Fermi-Dirac Distribution

$$f_F(E) = \frac{1}{1 + e^{\frac{E - E_F}{k_B T}}}$$

$$f_H(E) = 1 - f_F(E)$$

Built-In Potential

$$V_{bi} = \frac{k_B T}{e} \ln \left(\frac{N_A N_D}{n_i^2} \right)$$

Important Constants

$$e = 1.6 \times 10^{-19} \text{ C}$$

$$h = 6.64 \times 10^{-34} \text{ J} \cdot \text{s}$$

$$c = 3.0 \times 10^8 \text{ m/s}$$

$$hc = 1240 \text{ eV} \cdot \text{nm}$$

$$k_B = 8.617 \times 10^{-5} \text{ eV/K}$$

Physical Meaning of f_c and λ_c

The quantities

$$f_c$$

and

$$\lambda_c$$

represent the threshold values required for a photon to excite an electron across the semiconductor band gap. They correspond to the minimum photon energy necessary for absorption.

The band-gap energy is related to the photon energy by

$$E_g = hf_c = \frac{hc}{\lambda_c}$$

where

-

$$E_g$$

is the band-gap energy

-

$$f_c$$

is the critical (minimum) frequency

-

$$\lambda_c$$

is the critical (maximum) wavelength

Physical Meaning of f_c

$$f_c$$

is the smallest photon frequency capable of exciting an electron from the valence band into the conduction band.

Since photon energy is

$$E = hf$$

higher frequency corresponds to higher energy.

Therefore,

$$f < f_c$$

implies

$$hf < E_g$$

meaning the photon does not possess sufficient energy to overcome the band gap.

Physical Meaning of λ_c

$$\lambda_c$$

is the largest wavelength capable of exciting electrons across the band gap.
Since

$$E = \frac{hc}{\lambda}$$

larger wavelength corresponds to lower energy.
Therefore,

$$\lambda > \lambda_c$$

implies that the photon energy is too small to generate electron-hole pairs.

For the GaAs problem,

$$f_c \approx 3.42 \times 10^{14} \text{ Hz}$$

means photons with frequencies below this value cannot excite electrons across the band gap.
Similarly,

$$\lambda_c \approx 877 \text{ nm}$$

means wavelengths longer than 877 nm will not be strongly absorbed by the semiconductor.

Tutorials**Tutorial 1 - PART 1****Question 1**

If a charge is negative, the direction of the electric force on it is

- (a) In the same direction as the applied electric field
- (b) In the opposite direction as the applied electric field

Explain your reasoning.

Question 2

Consider a negatively charged particle which moves in an area of space where an electric field exists. No other forces act on the particle.

Which of the following is a correct statement? Explain your reasoning.

The particle

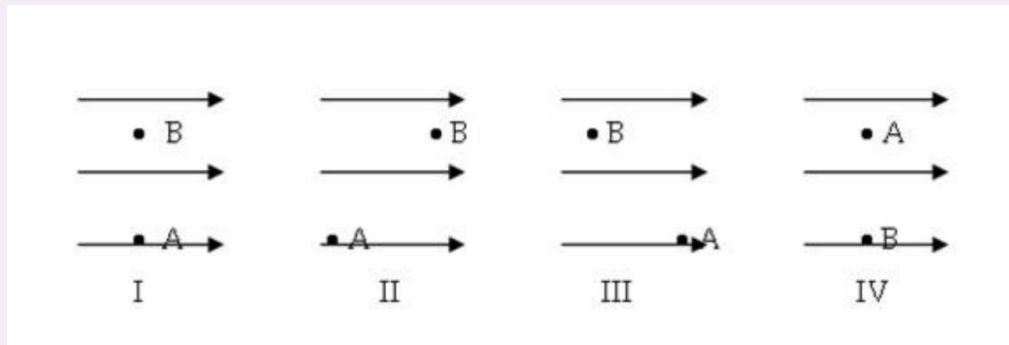
- (a) Gains potential energy and kinetic energy when it moves in the direction of the electric field
- (b) Loses electric potential energy when the particle moves in the direction of the electric field
- (c) Gains kinetic energy when it moves in the direction of the field
- (d) Gains electric potential energy when it moves in the direction of the field
- (e) Gains potential difference and electric potential energy when it moves in the direction of the field

Question 3

Four electrons move from point A to point B in a uniform electric field as shown below.

Rank the electrons in diagrams I through IV by their changes in potential energy from greatest to least when travelling from A to B .

Explain your reasoning.



Tutorial 1 Solutions - PART 1

Solution 1

Answer: (b) In the opposite direction as the applied electric field.

The electric force on a charge is given by

$$\vec{F} = q\vec{E}$$

For a negative charge ($q < 0$), the force vector points opposite to the electric field direction. Therefore, a negative charge experiences a force opposite to the applied electric field.

Solution 2

Answer: (b) Loses electric potential energy when the particle moves in the direction of the electric field.

For a negative charge,

$$\vec{F} = q\vec{E}$$

Since $q < 0$, the electric force acts opposite to the electric field.

When the particle moves in the direction of the electric field, the electric force acts opposite to the displacement.

Therefore, the electric field does negative work on the particle.

Using

$$\Delta U = -W_{\text{field}}$$

a negative value of work implies that the electric potential energy decreases.

Thus, the correct statement is (b).

Solution 3

The ranking of the changes in electric potential energy is

$$II > I = IV > III$$

The electric field points to the right, so the electric potential decreases in the direction of the field.

For an electron,

$$\Delta U = q\Delta V$$

where $q = -e$.

Diagram I and IV:

Points A and B are vertically aligned, so they are at the same electric potential.

Therefore,

$$\Delta V = 0 \Rightarrow \Delta U = 0$$

Diagram II:

Point B is to the right of A , so the electron moves in the direction of the electric field.

The electric potential decreases:

$$\Delta V < 0$$

Since the electron has negative charge,

$$\Delta U = q\Delta V > 0$$

Thus, Diagram II has the greatest increase in potential energy.

Diagram III:

In this case, point B is to the left of A , so the electron moves opposite to the electric field.

Therefore,

$$\Delta V > 0$$

and since $q = -e$,

$$\Delta U < 0$$

Diagram III has a larger horizontal displacement than Diagram IV, so it has the most negative change in potential energy.

Hence,

$$II > I = IV > III$$

Tutorial 2 - PART 1

Question 8

Explain the following concepts:

- Current carrier
- Hole
- p -type / n -type semiconductor
- Intrinsic semiconductor

Question 9

Suppose you know a material is p -type, and that the thermal equilibrium electron concentration is

$$n_0 = 2.0 \times 10^{10} \text{ cm}^{-3}$$

What can you say about the value of the intrinsic carrier concentration of this material?

Make the most precise statement you can.

Question 10

Consider the setup shown where the (conventional) current is flowing to the right through a semiconductor, and a magnetic field is pointing out of the page.

A voltmeter hooked up as shown reads a Hall voltage of +10 mV.

Is this an n -type or p -type material?

Explain carefully.

Tutorial 2 Solutions - PART 1

Solution 8

Current carrier:

A current carrier is a charged particle that contributes to electrical conduction in a material. In semiconductors, the current carriers are electrons and holes.

Hole:

A hole is the absence of an electron in the valence band. It behaves as an effective positive charge carrier.

p -type / n -type semiconductor:

A p -type semiconductor is formed by doping a semiconductor with acceptor atoms, producing holes as the majority carriers.

An n -type semiconductor is formed by doping with donor atoms, producing electrons as the majority carriers.

Intrinsic semiconductor:

An intrinsic semiconductor is a pure semiconductor with no intentional doping. Electrons and holes are present in equal concentrations.

Solution 9

For a semiconductor in thermal equilibrium,

$$n_0 p_0 = n_i^2$$

Since the material is p -type,

$$p_0 > n_0$$

Therefore,

$$n_i^2 = n_0 p_0 > n_0^2$$

Taking the square root gives

$$n_i > n_0$$

Hence,

$$n_i > 2.0 \times 10^{10} \text{ cm}^{-3}$$

Solution 10

The material is p-type.

The Hall effect arises because moving charge carriers experience a magnetic force:

$$\vec{F} = q\vec{v} \times \vec{B}$$

The conventional current flows to the right, and the magnetic field points out of the page.

For positive charge carriers (holes), the magnetic force direction produces the observed positive Hall voltage.

Since the measured Hall voltage is positive, the dominant charge carriers are positive.

Therefore, the semiconductor is *p*-type.

Tutorial 3 - PART 1

Topics: Band Structure, Fermi Energy, Fermi Function, Statistical Mechanics of Carrier Concentrations

Conceptual Questions

17. True or False: At any temperature, the Fermi energy is the energy level above which all electron states are empty.

18. (Based on Serway/Jewett – Objective Question 43.5)
 The conductivity of metals decreases with increasing temperature due to the increased frequency of electron collisions with atoms.
 In contrast, the conductivity of intrinsic semiconductors increases as T increases.
 What property of a semiconductor is responsible for this behavior?
 - (a) Collisions decrease as the temperature increases
 - (b) The number of free electrons and holes increases steeply with T
 - (c) The energy gap decreases with increasing temperature
 - (d) Electrons do not collide with atoms in a semiconductor

Quantitative Questions

19. (Based on Neaman, Problem 3.18)
 The energy gap of GaAs is 1.42 eV.
 - (a) Determine the minimum frequency f_c that an incident photon must have such that it can possibly elevate an electron to the conduction band.
 - (b) What is the corresponding wavelength λ_c of the photon you found in the previous part?
 - (c) Suppose the material is at $T = 0$ K and you shine light with $\lambda > \lambda_c$ on it.
 Describe what will happen in words.

Tutorial 3 Solutions - PART 1

Conceptual Question 17

Question

True or False: At any temperature, the Fermi energy is the energy level above which all electron states are empty.

Solution

False

At absolute zero ($T = 0\text{ K}$), all states above the Fermi energy are empty.

However, at temperatures above absolute zero, some electrons gain thermal energy and occupy states above the Fermi energy.

Therefore, the Fermi energy is not an absolute boundary between filled and empty states at all temperatures.

Conceptual Question 18**Question**

Why does the conductivity of intrinsic semiconductors increase with temperature?

Solution(b) The number of free electrons and holes increases steeply with T

As temperature increases, more electrons gain enough thermal energy to jump across the band gap from the valence band to the conduction band.

This creates:

more free electrons

and

more holes

which significantly increases conductivity.

Although collisions also increase with temperature, the dramatic increase in carrier concentration dominates in intrinsic semiconductors.

Quantitative Question 19(a)**Given**

$$E_g = 1.42\text{ eV}$$

Find the minimum photon frequency f_c .

Formula

$$E = hf$$

Therefore,

$$f_c = \frac{E_g}{h}$$

Convert electron volts to joules:

$$E_g = 1.42(1.6 \times 10^{-19})$$

$$E_g = 2.272 \times 10^{-19}\text{ J}$$

Using

$$h = 6.64 \times 10^{-34}\text{ J} \cdot \text{s}$$

we obtain

$$f_c = \frac{2.272 \times 10^{-19}}{6.64 \times 10^{-34}}$$

$$f_c = 3.42 \times 10^{14} \text{ Hz}$$

$$f_c = 3.42 \times 10^{14} \text{ Hz}$$

Quantitative Question 19(b)

Find the corresponding wavelength λ_c .

Formula

$$\lambda = \frac{c}{f}$$

Using

$$c = 3.0 \times 10^8 \text{ m/s}$$

and

$$f_c = 3.42 \times 10^{14} \text{ Hz}$$

we obtain

$$\lambda_c = \frac{3.0 \times 10^8}{3.42 \times 10^{14}}$$

$$\lambda_c = 8.77 \times 10^{-7} \text{ m}$$

Converting to nanometers:

$$\lambda_c = 877 \text{ nm}$$

$$\lambda_c = 877 \text{ nm}$$

Quantitative Question 19(c)

Question

Suppose the material is at $T = 0 \text{ K}$ and light with

$$\lambda > \lambda_c$$

is shined on the material.

Describe what happens.

Solution

If

$$\lambda > \lambda_c$$

then the photon energy is smaller than the band-gap energy because

$$E = \frac{hc}{\lambda}$$

and increasing wavelength decreases energy.

Therefore, the photons do not have enough energy to excite electrons from the valence band to the conduction band.

As a result:

- No electron-hole pairs are generated
- No significant absorption occurs

- The material remains non-conductive

Photons with $\lambda > \lambda_c$ cannot excite electrons across the band gap.

Tutorial 3 - PART 2

Question 20

(Neaman 4.11)

An n -type silicon sample has a resistivity of

$$5(\Omega \cdot \text{cm})$$

at

$$T = 300 \text{ K}$$

- What is the donor impurity concentration?
- What is the expected resistivity at

$$(i) T = 200 \text{ K}$$

$$(ii) T = 400 \text{ K}$$

Assume the doping is large so that

$$n_0 \gg n_i$$

You will need the mobility as a function of temperature.

Use

•

$$\mu_n = 2500 \frac{\text{cm}^2}{\text{V} \cdot \text{sec}} \quad \text{at } T = 200 \text{ K}$$

•

$$\mu_n = 1350 \frac{\text{cm}^2}{\text{V} \cdot \text{sec}} \quad \text{at } T = 300 \text{ K}$$

•

$$\mu_n = 700 \frac{\text{cm}^2}{\text{V} \cdot \text{sec}} \quad \text{at } T = 400 \text{ K}$$

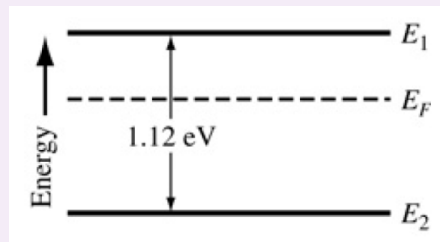
Question 21

(Based on Neaman 3.42)

Consider the energy levels shown in the figure.

Let

$$T = 300 \text{ K}$$



(a) If

$$E_1 - E_F = 0.30 \text{ eV}$$

determine:

- the probability that an energy state at

$$E = E_1$$

is occupied by an electron

- the probability that an energy state at

$$E = E_2$$

is empty

(b) Answer the same questions as the previous part, assuming

$$E_F - E_2 = 0.40 \text{ eV}$$

Question 22

Determine the probability that an allowed energy state is unoccupied by an electron if the state is below the Fermi energy level by

(a)

$$2k_B T$$

(b)

$$4k_B T$$

Question 23

(Based on Neaman, Problem 4.55)

Silicon at

$$T = 300 \text{ K}$$

is doped with acceptor atoms at a concentration of

$$N_a = 7.0 \times 10^{15} \text{ cm}^{-3}$$

(a) Determine

$$E_F - E_V$$

(b) Calculate the concentration of additional acceptor atoms that must be added to move the Fermi level a distance

$$k_B T$$

closer to the valence band edge.

Tutorial 3 Solutions - PART 2

Solution 20**Given**

$$\rho = 5 \, \Omega \cdot \text{cm}$$

$$\mu_n = 1350 \frac{\text{cm}^2}{\text{V} \cdot \text{s}}$$

(a) Donor Impurity ConcentrationFor an n -type semiconductor,

$$\sigma = en\mu_n$$

and

$$\rho = \frac{1}{\sigma}$$

Therefore,

$$n = \frac{1}{e\rho\mu_n}$$

Substitute the values:

$$n = \frac{1}{(1.6 \times 10^{-19})(5)(1350)}$$

$$n = 9.26 \times 10^{14} \text{ cm}^{-3}$$

Thus,

$$\boxed{N_D \approx 9.3 \times 10^{14} \text{ cm}^{-3}}$$

(b) Resistivity at Different Temperatures

Since

$$\rho = \frac{1}{en\mu_n}$$

and the carrier concentration remains approximately constant,
we only change the mobility.**At $T = 200 \text{ K}$**

$$\mu_n = 2500 \frac{\text{cm}^2}{\text{V} \cdot \text{s}}$$

$$\rho_{200} = \frac{1}{(1.6 \times 10^{-19})(9.26 \times 10^{14})(2500)}$$

$$\boxed{\rho_{200} \approx 2.7 \, \Omega \cdot \text{cm}}$$

At $T = 400 \text{ K}$

$$\mu_n = 700 \frac{\text{cm}^2}{\text{V} \cdot \text{s}}$$

$$\rho_{400} = \frac{1}{(1.6 \times 10^{-19})(9.26 \times 10^{14})(700)}$$

$$\boxed{\rho_{400} \approx 9.6 \, \Omega \cdot \text{cm}}$$

Solution 21

Use the Fermi-Dirac distribution:

$$f(E) = \frac{1}{1 + e^{(E-E_F)/k_B T}}$$

At

$$T = 300 \text{ K}$$

$$k_B T = 0.026 \text{ eV}$$

(a) Probability that $E = E_1$ is occupied

Given

$$E_1 - E_F = 0.30 \text{ eV}$$

Then

$$f(E_1) = \frac{1}{1 + e^{0.30/0.026}}$$

$$f(E_1) = 9.75 \times 10^{-6}$$

Thus,

$$P(E_1 \text{ occupied}) = 9.75 \times 10^{-6}$$

Probability that $E = E_2$ is empty

The hole probability is

$$1 - f(E)$$

Since E_2 is equally far below E_F ,

$$P(E_2 \text{ empty}) = 9.75 \times 10^{-6}$$

$$P(E_2 \text{ empty}) = 9.75 \times 10^{-6}$$

(b) If

$$E_F - E_2 = 0.40 \text{ eV}$$

then

$$P(E_2 \text{ empty}) = \frac{1}{1 + e^{0.40/0.026}}$$

$$P(E_2 \text{ empty}) = 2.08 \times 10^{-7}$$

$$P(E_2 \text{ empty}) = 2.08 \times 10^{-7}$$

Solution 22

The probability that a state is unoccupied is

$$f_H(E) = 1 - f_F(E) = \frac{1}{1 + e^{(E_F - E)/k_B T}}$$

(a) State below E_F by $2k_B T$

$$E_F - E = 2k_B T$$

Then

$$f_H(E) = \frac{1}{1 + e^2}$$

$$f_H(E) = 0.119$$

Therefore,

$$P(\text{unoccupied}) = 11.9\%$$

(b) State below E_F by $4k_B T$

$$E_F - E = 4k_B T$$

Then

$$f_H(E) = \frac{1}{1 + e^4}$$

$$f_H(E) = 0.018$$

Therefore,

$$P(\text{unoccupied}) = 1.8\%$$

Solution 23

Given

$$N_A = 7.0 \times 10^{15} \text{ cm}^{-3}$$

$$T = 300 \text{ K}$$

For silicon,

$$N_V = 1.04 \times 10^{19} \text{ cm}^{-3}$$

(a) Determine $E_F - E_V$

Use

$$E_F - E_V = k_B T \ln \left(\frac{N_V}{p_0} \right)$$

Since the semiconductor is p -type,

$$p_0 \approx N_A$$

Thus,

$$E_F - E_V = (0.026) \ln \left(\frac{1.04 \times 10^{19}}{7.0 \times 10^{15}} \right)$$

$$E_F - E_V = 0.19 \text{ eV}$$

Therefore,

$$E_F - E_V = 0.19 \text{ eV}$$

(b) Additional Acceptor Concentration

To move the Fermi level closer to the valence band by

$$k_B T$$

the acceptor concentration must increase by a factor of

$$e$$

Therefore,

$$N'_A = e N_A$$

$$N'_A = (2.718)(7.0 \times 10^{15})$$

$$N'_A = 1.90 \times 10^{16} \text{ cm}^{-3}$$

Additional acceptor atoms needed:

$$\Delta N_A = N'_A - N_A$$

$$\Delta N_A = 1.90 \times 10^{16} - 7.0 \times 10^{15}$$

$$\Delta N_A = 1.2 \times 10^{16} \text{ cm}^{-3}$$

Thus,

$$\boxed{\Delta N_A = 1.2 \times 10^{16} \text{ cm}^{-3}}$$

Tutorial 3 - PART 3**Question 24**

An n -type silicon sample has a resistivity of

$$5(\Omega \cdot \text{cm})$$

at

$$T = 300 \text{ K}$$

What is the Fermi energy position relative to the conduction band?

Given:

$$\mu_n = 1350 \frac{\text{cm}^2}{\text{V} \cdot \text{sec}}$$

at

$$T = 300 \text{ K}$$

Question 25

The Fermi energy of an n -type silicon sample is between

$$E_C$$

and the middle of the energy gap.

When

moves toward

E_F

E_C

will the resistivity of the sample increase or decrease?
Explain your reasoning.

Tutorial 3 Solutions - PART 3

Solution 24**Given**

$$\rho = 5 \, \Omega \cdot \text{cm}$$

$$\mu_n = 1350 \frac{\text{cm}^2}{\text{V} \cdot \text{s}}$$

For an n -type semiconductor:

$$\sigma = en\mu_n$$

and

$$\rho = \frac{1}{\sigma}$$

Thus,

$$n = \frac{1}{e\rho\mu_n}$$

Substitute values:

$$n = \frac{1}{(1.6 \times 10^{-19})(5)(1350)}$$

$$n = 9.26 \times 10^{14} \text{ cm}^{-3}$$

Use

$$E_C - E_F = k_B T \ln \left(\frac{N_C}{n_0} \right)$$

For silicon at 300 K:

$$N_C = 2.8 \times 10^{19} \text{ cm}^{-3}$$

and

$$k_B T = 0.026 \text{ eV}$$

Therefore,

$$E_C - E_F = 0.026 \ln \left(\frac{2.8 \times 10^{19}}{9.26 \times 10^{14}} \right)$$

$$E_C - E_F = 0.267 \text{ eV}$$

Thus,

$$E_F \text{ is } 0.267 \text{ eV below } E_C$$

Solution 25

When the Fermi level moves closer to the conduction band:

$$E_F \rightarrow E_C$$

the electron concentration increases.

From semiconductor statistics:

$$n_0 = N_C e^{-(E_C - E_F)/k_B T}$$

As

$$E_C - E_F$$

becomes smaller, the exponential term becomes larger, so the number of free electrons increases.

Conductivity is

$$\sigma = en\mu_n$$

Thus, increasing electron concentration increases conductivity.

Since

$$\rho = \frac{1}{\sigma}$$

the resistivity decreases.

The resistivity decreases as E_F moves closer to E_C

Practice Problems

Set 6

Question 1

True or False:

The formulas

$$n_0 = N_c e^{-(E_c - E_F)/k_B T}$$

and

$$p_0 = N_v e^{-(E_F - E_v)/k_B T}$$

do not include the effects of doping.

Explain your answer.

Solution 1

Answer: False

The effects of doping are included indirectly through the Fermi energy level E_F .

When donor or acceptor atoms are added to a semiconductor, the Fermi level shifts:

- In n -type material, E_F moves closer to the conduction band.
- In p -type material, E_F moves closer to the valence band.

Since the carrier concentration formulas depend on E_F , doping changes both n_0 and p_0 . Therefore, the statement is false.

Question 2

When donor atoms are introduced into a semiconductor sample, why does the equilibrium hole concentration decrease?
Explain in words.

Solution 2

When donor atoms are added, extra electrons are introduced into the semiconductor. These electrons recombine with holes, causing previously unoccupied states in the valence band to become occupied.

As a result, the equilibrium hole concentration decreases.

This is also consistent with the mass-action law:

$$n_0 p_0 = n_i^2$$

As the electron concentration n_0 increases, the hole concentration p_0 must decrease so that the product remains constant.